Sample Papers

	MIDTERM EXAMINATION SPRING 2007 CS402 - THEORY OF AUTOMATA (Session - 4)	Marks: 35 Time: 90min

StudentID/LoginID: _____

Student Name: _____

Center Name/Code: _____

Exam Date: Saturday, April 28, 2007

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1. **This examination is closed book, closed notes, closed neighbors.**
2. **Answer all questions.**
   a. **There is no choice.**
   b. **You will have to answer correctly all questions in this examination to get the maximum possible marks.**
3. **Do not ask any questions about the contents of this examination from anyone.**
   a. **If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.**
   b. **If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.**

You are allowed to use only MS Word for drawing Diagrams and Symbols.

****WARNING: Please note that Virtual University takes serious note of unfair means. Anyone found involved in cheating will get an `F` grade in this course.**

For Teacher's use only

Question	1	2	3	4	5	6	7	8	9	10	Total
Marks											
Question	11	12	13	14	15	16	17	18			
Marks											
Question											
Marks											

Question No: 1 (Marks: 1) - Please choose one

Length of a null string is supposed to be 1.

- ▶ True
- ▶ False

Question No: 2 (Marks: 1) - Please choose one

There is no difference between Word and String

- ▶ True
- ▶ False

Question No: 3 (Marks: 1) - Please choose one

There may be two RE representing the same language.

- ▶ True
- ▶ False

Question No: 4 (Marks: 1) - Please choose one

NFA – null can be considered as TG and vise versa.

- ▶ True
- ▶ False

Question No: 5 (Marks: 1) - Please choose one

$\sum$ = {aΛ, b}, length(aAbaAaAb) = 5.

- ▶ True
- ▶ False

Question No: 6 (Marks: 1) - Please choose one

If s=abcd is a string defined over $\sum$ = {a,b,c,d} then reverse of s is dcba.

- ▶ True
- ▶ False

Question No: 7 (Marks: 1) - Please choose one

The language equal means number of a's and b's are equal with no null string.

- ▶ True
- ▶ False

Question No: 8 (Marks: 1) - Please choose one

Palindrome is a regular language.

- ▶ True
- ▶ False

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Question No: 9 (Marks: 1) - Please choose one

If s = babab then palindrome of s = rev(s).

- ▶ True
- ▶ False

Question No: 10 (Marks: 1) - Please choose one

TG must have only one start state.

- ▶ True
- ▶ False

Question No: 11 (Marks: 1) - Please choose one

If a language can be accepted by FA then it can be accepted by a TG as well.

- ▶ True
- ▶ False

Question No: 12 (Marks: 1) - Please choose one

Length of output string is 1 less then that of input string is mealy machine.

- ▶ True
- ▶ False

Question No: 13 (Marks: 1) - Please choose one

Formal languages are called Semantic languages.

- ▶ True
- ▶ False

Question No: 14 (Marks: 1) - Please choose one

Every NFA can be converted to an FA

- ▶ True
- ▶ False

Question No: 15 (Marks: 1) - Please choose one

In mealy machine output character are mentioned on the transition.

- ▶ True
- ▶ False

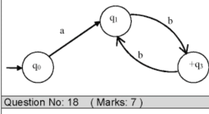
Question No: 16 (Marks: 8)

Differentiate between the following terms:

- **Mealy and Moore machine.**
- **NFA and FA.**

Question No: 17 (Marks: 5)

Build an FA corresponding to NFA given below.



Question No: 18 (Marks: 7)

Build an NFA equivalent to $FA_1 \cup FA_2$, where FA_1, FA_2 are given below.

FA_1



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FA_2

